BiTeM at TRECVID 2026 VQA: What a 500-Item Development Set Could Not See

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Abstract

We entered both TRECVID 2026 VQA tasks with one locally hosted, open-weight pipeline built on Qwen3-VL and Qwen3.5, and placed 7th of 15 teams on answer generation and 10th of 13 on multiple choice. Our development set could not rank our own runs: the multiple-choice run we nominated as primary finished fourth of our five, while the 122B run, entered for diversity, and a fusion run we nearly left out finished first and second, 3.5 and 3.1 points higher. Four identical runs on our 497 development items spread over 4 points, and the 3.5-point effect we missed needs about 900 items to be detected reliably. We also audit the 2025 collection, our development data. Its multiple-choice gold is the free-text reference copied verbatim, a few golds are non-answers or unreachable, and the longest option is correct on 47 % of its four-option questions against 25 % chance (41 % against 15 % on the 2026 labels). Finally, the official answer-generation measures disagree: the same run ranks 14th, 20th and 41st of 44 depending on the measure, and a slate of one real answer and nine nonsense strings outscores every official run on METEOR NDCG@10.

Keywords

video question answering, multiple choice, benchmark artifacts, run-to-run variance, evaluation measures

Team Name

BiTeM

Subtasks

Task 1: Answer Generation (AG)
Task 2: Multiple Choice (MC)

Structured Abstract

Task 1, Answer Generation. Training data. None: no model was trained or fine-tuned. Pretrained Qwen3-VL-8B answers from a pool of 16 frames per video. The released 2025 VQA collection [2] served as development data and as the source of the few-shot examples. Speech is transcribed with faster-whisper large-v3-turbo, and candidate answers are clustered with all-MiniLM-L6-v2 sentence embeddings.

Runs.

- `BiTeM_cons8b_s8`: eight independent samples of a single short answer, each on a random 8 of the 16 frames, clustered and ranked by how often an answer recurs.
- `BiTeM_cons8b_s16`: the same with 16 samples; our flagship.
- `BiTeM_fewshot8b`: eight samples, plus the four nearest 2025 (question, reference) pairs as examples.
- `BiTeM_all8b`: as `fewshot8b`, plus a speech transcript for questions a lexical router flags as verbal.
- `BiTeM_v3control`: our earlier prompt format, one call for ten increasingly speculative answers, entered as a control.

Significant differences. The four sampling runs span 0.005 STS@1. The control is last of our five on STS@1, 0.023 below the next run, but first on METEOR@1 (0.036 above the next) and on METEOR NDCG@10 (0.077 above the next).

Component contributions. Asking for one short answer many times, instead of ten ranked answers in one call, gained 0.07 STS on development topics and cost METEOR. Doubling the samples, adding examples and adding routed transcripts each moved official STS@1 by 0.005 or less.

Lessons. The official measures disagree about which of our runs is best. After submission, minimum-Bayes-risk selection over 16 to 128 samples improved only the measure it used as its utility (Figure 7).

Task 2, Multiple Choice. Training data. None: nothing was trained or fine-tuned. Pretrained Qwen3-VL-8B, 30B-A3B and 32B, and Qwen3.5-122B, all served by Ollama. Development used 497 eight-option items built from the 2025 collection [2]. Transcripts are produced as in Task 1, for routed questions only.

Runs (short names as in Table 2).

- `BiTeM_mc32b` (`mc_32b_prod`): Qwen3-VL-32B ranks every option from the frame pool, with a transcript when routed; nominated primary.
- `BiTeM_mc30b` (`mc_30b_prod`): the same with Qwen3-VL-30B-A3B.
- `BiTeM_mc122b` (`mc_122b_prod`): the same with Qwen3.5-122B.
- `BiTeM_mc32bdense` (`mc_32b_dense`): `mc32b` with 32–64 frames for videos longer than 40 s.
- `BiTeM_mcfusion` (`mc_fusion`): reciprocal-rank fusion of the 32B, 122B and 30B option rankings.

Significant differences, by exact McNemar test against `mc32b`: `mc122b` +3.5 points and `mcfusion` +3.1, both at $p < 10^{-5}$ and level with each other ($p = 0.46$); `mc32bdense` +0.1 ($p = 0.89$); `mc30b` −1.5 ($p = 0.051$).

Component contributions. The choice of model and the three-model fusion each gained about 3 points. Denser frames on long clips gained nothing measurable.

Lessons. Our 497 development items could not detect 3-point differences, and we read that as evidence of no difference. The released options carry a length artifact as strong in 2026 as in 2025. A cascade that escalates likely 8B errors to 32B was built and withheld: it agreed with `mc32b` on 96 % of top-1 answers and cost more per question than running 32B everywhere.

1 Introduction

The TRECVID Video Question Answering track [2] asks systems to answer natural-language questions about short videos. Task 1, Answer Generation (AG), takes ten distinct ranked free-text answers per question. Task 2, Multiple Choice (MC), takes a ranking of the released options, scored by exact string match against one correct option.

Our system is standard: an open-weight video-language model sampled on random subsets of a video's frames, with answers aggregated across samples and a speech transcript added when a question seems to need one. Everything runs locally, and no commercial API is called. Our best runs placed 7th of 15 teams on Task 1 and 10th of 13 on Task 2.

Most of what we learned this year is about evaluation: how small a difference 500 items can detect, and what this task's labels and measures reward. Our development set could not rank our own Task 2 runs. The run we nominated as primary finished fourth of our five, while the 122B run, entered for diversity, and a fusion run we nearly left out finished first and second. Our contributions are:

- a replicate study of what 497 development items can resolve: four runs of one configuration spread over 4 points, a fifth of the items flip between runs, and two 3-point effects we dismissed on this set were real on the 2,428 official questions;
- an audit of the MC labels: the gold option is the AG reference copied verbatim, six golds are non-answers, four cannot be matched by any run, and the longest option is correct far above chance in both years;
- evidence that the official AG measures disagree and can be gamed: one run ranks from 14th to 41st of 44 depending on the measure, a selection rule improves only the measure it optimises, and one answer padded with nine nonsense strings tops every official run on METEOR NDCG@10;
- a list of issues for the organisers, each with a fix (Table 5).

2 Related Work

Partial-input baselines and option length. Benchmarks can often be solved without their input. Gururangan et al. [12] and Poliak et al. [25] showed this for NLI, and Kaushik and Lipton [17] and Sugawara et al. [30] for reading comprehension. For multiple choice, showing only the options beats the majority baseline in 11 of 12 datasets [3]. Our longest-option baseline reads neither the video nor the question. The length cue is known in educational testing: Millman et al. [22] described test-wiseness, and Haladyna et al.'s item-writing guidelines [13] ask for options of about equal length. ReClor [35] used such biases to split its test set into easy and hard halves.

How benchmarks acquire artifacts. Adversarial filtering [18, 36] and balanced construction [11] are the usual remedies. Aroyo and Welty [1] made explicit the assumptions behind single-annotator gold, Pavlick and Kwiatkowski [24] showed that annotator disagreement does not average out, and Northcutt et al. [23] measured what label errors cost downstream.

Video QA and single-frame bias. About half of MovieQA can be answered without the video [14]. TVQA [20] and NExT-QA [34] target temporal and causal reasoning, while Buch et al. [4] and Lei et al. [19] found that much of video QA needs little temporal information. Video-MME [9] made duration an explicit axis, which matters here because a fifth of the 2026 questions come from clips longer than 40 seconds.

Variance and power. Random seeds alone produce differences significant at $p < 10^{-4}$ [27], typical NLP experiments are underpowered [5], and LLM results move with prompt format [28] and option order [37]. We compare paired runs with the exact McNemar test [21], which Dietterich [8] recommends for algorithms run once.

Abstention and cascades. Classification with a reject option goes back to Chow [7], and selective classification for deep networks to Geifman and El-Yaniv [10]. SQuAD 2.0 [26] made abstention part of a QA task, Kadavath et al. [15] asked models whether they know the answer, and Wen et al. [33] survey the area. Like Kamath et al. [16], we train a separate error predictor. Post-abstention [31], cascades [6] and self-consistency [32] are the closest designs. RFC 4180 [29] specifies the CSV quoting whose violation made four golds unreachable.

3 Methods

3.1 Tasks and data

Scoring. Task 1 is scored with STS, METEOR and BERTScore F1. Before submission we took the score to be each measure's mean over the ten answers, as in the 2025 scorer. The 2026 report gives, for each measure, the best answer and the mean answer within the top 1, 5 and 10 and an NDCG computed within the slate, and its summary uses the three @1 scores and the three NDCG@10. Task 2 is scored by top-1 accuracy and MRR under exact string matching, so a re-typed or trimmed option counts as wrong.

The 2026 collection. Both tasks share 2,443 questions over 1,956 videos. In Task 2, 1,971 questions have eight options and 472 have four, for a chance rate of 14.9 %. The MC file was released twice, and 2,435 of its option lists changed between the two versions, so we rebuilt every run against

the second. Video durations are bimodal: the median is 10 seconds, but 212 clips last longer than 40 seconds (up to 389) and carry 479 questions, a fifth of the task. No development video is longer than 30 seconds.

Development data. The 2025 test collection [2] was released with its ground truth, the same file the public scorer uses, so the 2025 evaluation can be reproduced exactly. We use its 1,978 scoreable questions: 18 have no video and 4 have an unreachable gold (Section 4.2). For Task 2 we first used the first 300 questions of the file. That slice turned out to be the most 8B-favouring window of its size in the whole collection, so we replaced it with 497 questions sampled to the 2026 mix of question types and padded from four to eight options with length-matched distractors from other questions. This set ranks systems against each other but does not predict 2026 accuracy: its options are less alike than those of the real eight-option items (mean token overlap 0.07 against 0.18).

3.2 System

Both systems use a pool of 16 evenly spaced frames per video, an optional speech transcript, and a model served locally by Ollama: Qwen3-VL at 8B, 30B-A3B or 32B, or Qwen3.5-122B. Nothing was trained or fine-tuned. Figure 1 shows the two pipelines, and Figure 2 follows one question through the first.

Task 1. We sample S answers ($S = 8$ or 16), each from its own random 8 of the 16 frames, with a prompt that asks for a single short answer in the style of the references: one clause, no hedging, a pronoun for a single person, number words for counts. Two additions are optional: a transcript, for questions a lexical router flags as verbal, and the four most similar 2025 (question, reference) pairs as examples. Answers are clustered with sentence embeddings and ranked by how often they recur, and surface variants of the top answer fill the ten required slots. The prompt format was our largest Task 1 effect. Asking for one answer many times, instead of ten ranked answers in one call "with increasingly speculative alternatives", gained 0.070 STS on development topics (2.7 standard errors), but lost 0.03 METEOR and 0.08 METEOR NDCG@10. Figure 2 shows why sampling over frame subsets helps: a sample can only read what its frames show, and the answers that recur are the ones most frames support.

Task 2. Each question gets three votes. A vote shows the model a random subset of 8, 12 or 16 frames (more for questions whose wording suggests text, audio or timing), the question, a transcript when the question seems to concern audio (14 % of test questions), and the options lettered in released order. The model describes the evidence, evaluates each option and ends with a full ranking. The first vote runs at temperature 0 and the other two at 0.7. Votes are combined by MRR-weighted Borda count; a vote that names fewer than two options is asked again once and otherwise dropped.

Two-tier escalation. The gold is always among the released options, so an injected option "None of the above can be determined from the frames shown" can never be correct, but it can signal doubt. Offered in the answering pass, it costs accuracy: 61.0 % against a mean of 67.6 % for the 8B runs without it. Its flag is informative, though: items where it fires are wrong 63 % of the time, against 22 % for the rest. We therefore keep it out of the answers. A clean 8B pass answers. A second 8B pass with the injected option supplies features (the abstention flag, the Borda margin between the top two options, disagreement between votes, option count), and a logistic regression on them sends the items most likely to be wrong to a 32B pass. We fixed this policy in writing before the 2026 run that fed it had finished.

3.3 Runs

Every run passes the official validator on all 2,443 questions. Tables use the short names given in parentheses.

Task 1, all Qwen3-VL-8B. **BiTeM_cons8b_s8**: the sampling pipeline with $S = 8$. **BiTeM_cons8b_s16**: the same with $S = 16$; our flagship. **BiTeM_fewshot8b**: $S = 8$ plus four 2025 examples. **BiTeM_all8b**: as `fewshot8b`, plus routed transcripts. **BiTeM_v3control**: our earlier prompt format, one call for ten answers, entered as a control we expected to be our worst run.

Task 2. **BiTeM_mc32b** (`mc_32b_prod`): Qwen3-VL-32B, nominated as primary. **BiTeM_mc30b** (`mc_30b_prod`): Qwen3-VL-30B-A3B. **BiTeM_mc122b** (`mc_122b_prod`): Qwen3.5-122B. **BiTeM_mc32bdense** (`mc_32b_dense`): `mc32b` with 32–64 frames for the 212 videos longer than 40 seconds and 16 frames per vote. **BiTeM_mcfusion** (`mc_fusion`): reciprocal-rank fusion of the 32B, 122B and 30B option rankings. We entered `mc30b` and `mc122b` for diversity; `mc30b` disagrees with `mc32b`'s top answer on 20 % of questions. We withheld four other runs: a plain 8B run; an 8B run with partial transcripts, which agrees with the plain one on 98 % of top answers and so cannot differ from it by more than 2.1 points; the two-tier run, dropped by a rule fixed in writing before it was built (it agreed with `mc32b` on 96.0 % of top answers, above our 95 % limit); and a hybrid of the dense and fusion runs.

4 Results

The first five subsections report development measurements on 2025 data, made before the official scores were released. The last two use the official 2026 scores.

4.1 How much the development set can resolve

We ran one 8B configuration four times on the 497 development items, with the same model, prompt, frames and settings. The runs differed only in inert command-line details, the frame-sampling seed and the GPU partition that served them, and Ollama does not seed sampling at temperature 0.7. They scored 65.6, 67.4, 67.8 and 69.6 % (Figure 3a).

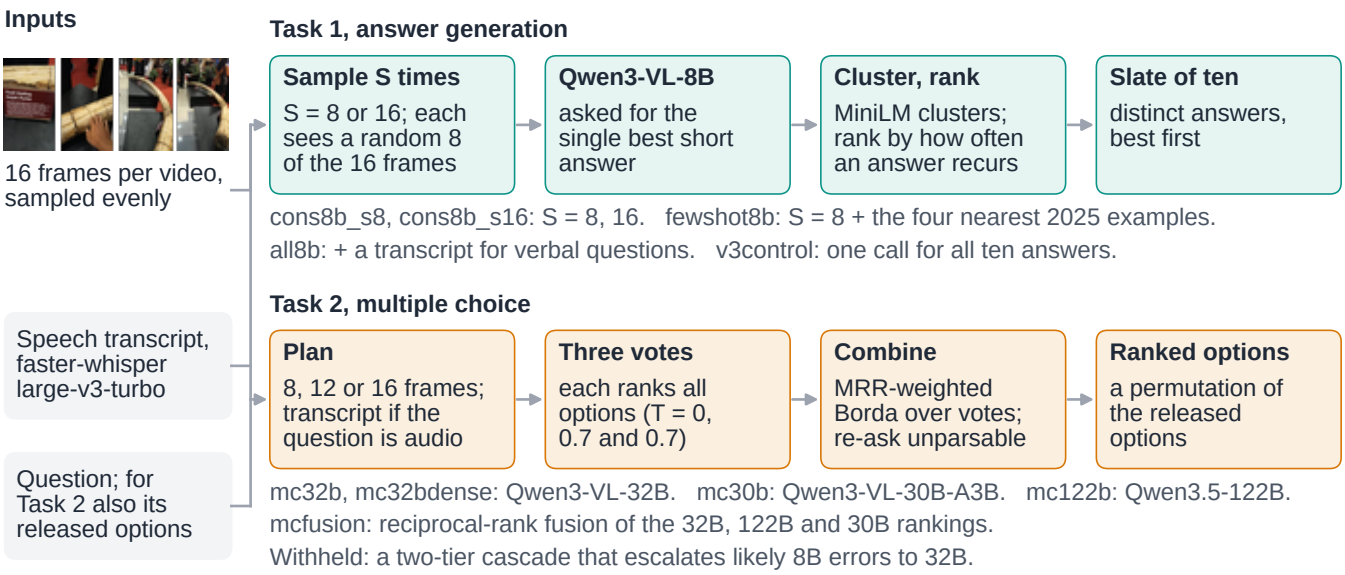


Figure 1: The two pipelines. Both tasks share one pool of 16 evenly spaced frames per video, an optional speech transcript and a locally served model. Task 1 asks for one short answer S times, each time on a random 8 of the 16 frames, and keeps what recurs; Task 2 casts three votes that each rank every option, and combines them. The notes under each lane name the variant behind each submitted run.

One of the six exact McNemar tests between pairs of these runs gives $p = 0.017$. That is not a fault of the test, which already accounts for items that flip from run to run, and one rejection in six tests is unremarkable. The 4-point range is not a usable threshold either, since the range of k runs grows with k. The informative quantity is the variance. A fifth of the items (20.7 %) flip between the four runs. If flips were independent across items, run accuracy would vary with a standard deviation of 1.07 points; the observed one is 1.65, which implies a component of about 1.3 points shared by all items of a run. This is the part of run-to-run variation that McNemar's test cannot see, because it treats items as independent. With four runs the excess is suggestive, not established (permutation test within items, $p = 0.07$), and the four 32B runs show little of it (1.10 observed against 0.98 predicted, $p = 0.30$).

Comparisons between single runs inherit this variation. The same 32B run is between 2.4 and 6.4 points above the four 8B runs, depending on which replicate is the baseline, and its McNemar p ranges from 0.19 to 0.0002. Pooling each item's replicates gives the comparisons the set can support. Four 32B runs beat four 8B runs by 3.6 points (sign-flip permutation over items, $p = 0.006$; adding the shared run-level component estimated above leaves $p \approx 0.01$). Two Qwen3.5-122B runs beat the four 32B runs by 1.8 points ($p = 0.16$), a difference the set cannot separate from zero; the official set later put it at 3.5 points ($p < 10^{-5}$, Section 4.6). At the discordance between our replicates, 497 items give 80 % power only to differences of about 4.3 points. An oracle that counts an item as solved when any of the four 8B runs is right scores 78.1 %, 10.5 points above their mean, so much of the apparent complementarity between our runs is the same flipping counted several times.

4.2 What the multiple-choice labels are made of

MC is AG with distractors. In the 2025 collection, every MC gold option is identical to the AG reference for its question, in all 1,996 items that have both. There is one reference per question and a constant relevance grade. The two tasks therefore share their annotation errors: a wrong reference costs points on Task 1 and becomes the correct option on Task 2, and tuning on one task while reporting on the other is not cross-validation.

Non-answers and contested items. Six references are non-answers ("I don't know." four times, "Nothing." and "None."), and all six became MC golds. Q40 asks which chemical is being mixed and keys "I don't know." over "Sodium bicarbonate.", so a system that names the chemical is wrong on both tasks. Reading 22 contested items, we also found keys with several true options, golds the video does not support, factually wrong golds, and questions with a false premise (Q135 asks about a woman in pink, and there is none). These items were chosen for being contested, so they do not estimate a defect rate.

The option-length artifact. Choosing the longest option, without the video or the question, is correct on 47.1 % of the 2025 four-option questions, against 25 % chance (Figure 4).

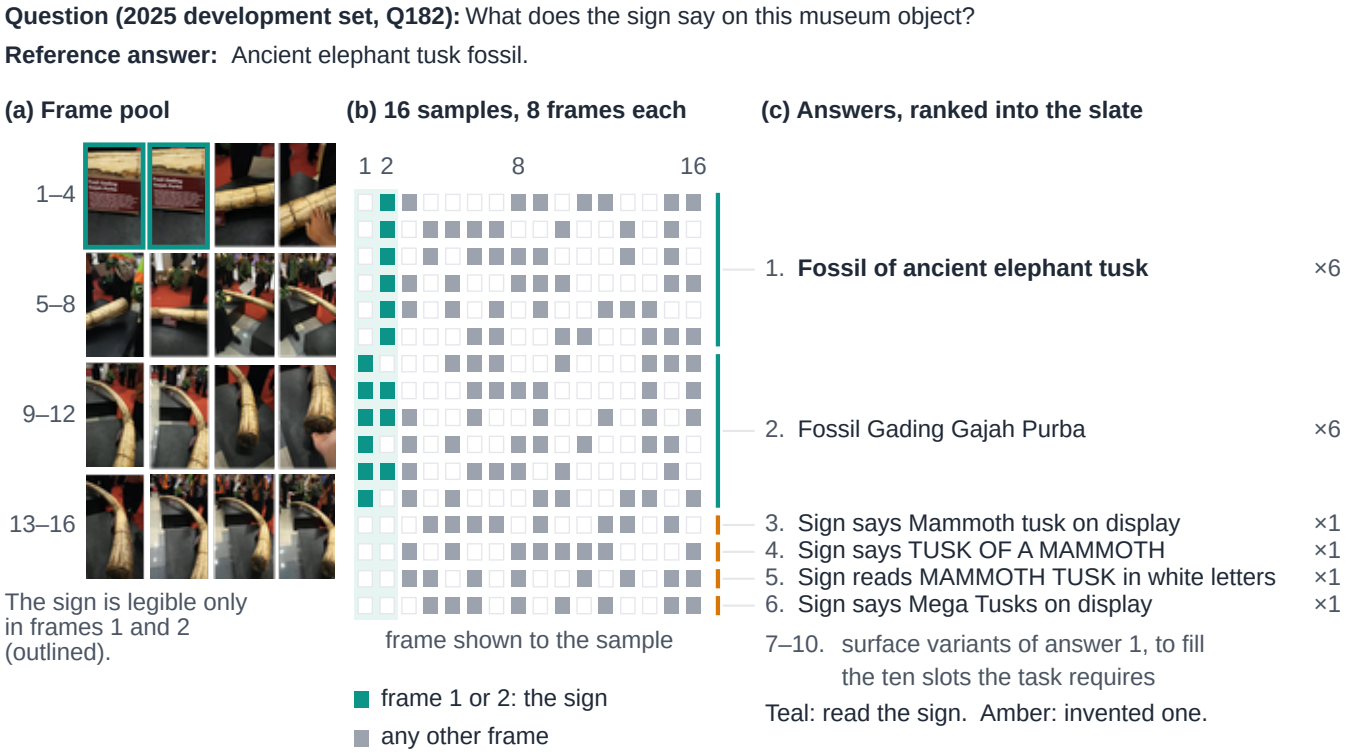


Figure 2: One development question through the Task 1 pipeline, recomputed from the run's own records. (a) The frame pool; the sign is legible only in frames 1 and 2. (b) The eight frames each of 16 samples was shown, rows grouped by the answer the sample gave. (c) Those answers in slate order, with how many samples gave each. Every sample shown frame 1 or 2 read the sign, copying it when frame 1 was among its frames and translating it when only frame 2 was; the four that saw neither each invented a different sign. Counting repeats puts the translation, which matches the reference, first.

The effect depends on the question type. It is strongest for *how*, *why* and *what* questions (54 to 63 %), where the gold is a sentence written while watching and the distractors were written afterwards to be wrong. It disappears or reverses for counting and yes/no questions, whose answer space is closed. On the released 2026 labels the same rule is correct on 40.9 % of questions. Corrected for chance, as (acc − chance)/(1 − chance), that is 0.30, against 0.29 in 2025, and on four-option items it is 49.2 % against 47.1 %: the artifact has not weakened. Our best 2025 development score, 74.8 %, is 28 points above this baseline.

Unreachable golds. Four 2025 golds (Q332, Q387, Q596, Q1745) do not appear among their own options, so no run could score them. The AG topics file puts a space before each opening quote, which breaks CSV quoting [29] and splits 41 rows at internal commas; in three of them the split reaches the answer column. Q332 has a missing quote in the MC file. Together with the 18 questions whose videos were unavailable, which the 2025 scorer skips but keeps in its denominator, these cap the official 2025 MC score at 98.9 %. A five-line check (each row parses to its declared fields, and each gold appears among its options) names all four before release.

4.3 Development ablations

Table 1 lists every MC effect we measured on development data, and Figure 5 plots the whole-set ones against the range of the four identical runs. Besides model size (Section 4.1), two effects are clearly larger than run-to-run variation, and both act on a subset of questions.

Transcripts. On the 48 development questions that receive a transcript, 32B accuracy rises from 52.1 to 72.9 % ($p = 0.013$), which is 10 more correct answers, or 2.0 points over all 497 items. The whole-set comparison shows 2.6 points because its transcript runs also used a larger frame budget. Routing transcripts only to questions about speech scored the same as giving one to every question that has one (74.65 % both, 14 discordant items each way), so we kept routing because it is cheaper. Sixteen frames with a transcript beat eight frames without one by 10 points on the audio slice ($p = 0.012$), two thirds of it from the transcript.

Frames. Sixteen frames instead of eight changed nothing on the temporal or audio slices (+0.0 and −0.8 points, 120 items each). No development video is longer than 30 seconds, while a fifth of the test questions come from clips of up to

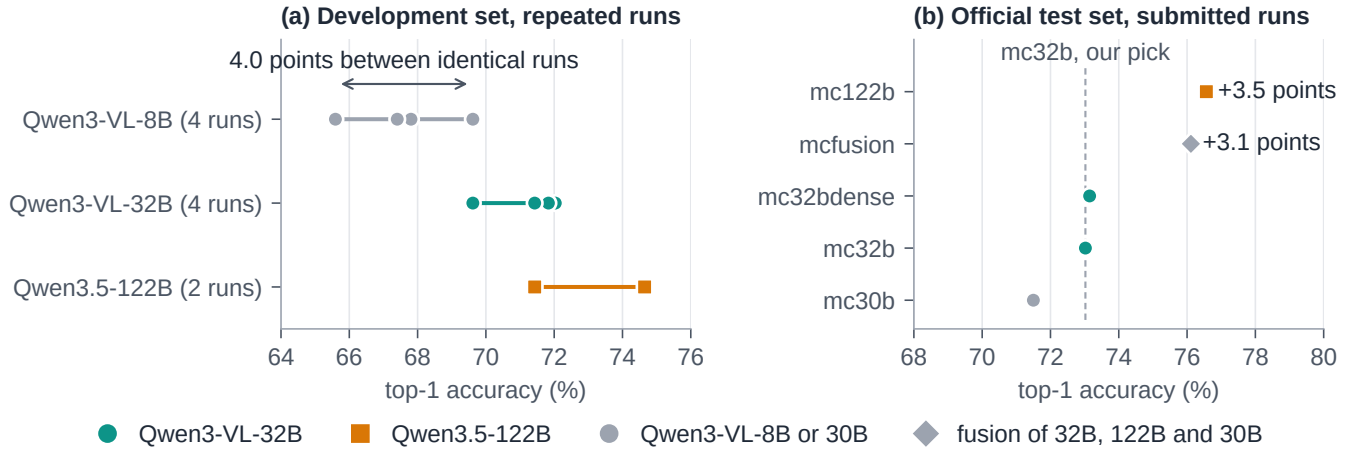


Figure 3: Task 2: what the development set could and could not resolve. (a) Top-1 accuracy of repeated runs on the 497-item development set. The four 8B runs share one configuration and still span 4.0 points, and the 32B and 122B ranges overlap. (b) Official top-1 of the five submitted runs on 2,428 questions, with the difference from `mc32b`, the run we nominated as primary. Both labelled differences are significant by exact McNemar test ($p < 10^{-5}$). The panels share one scale, so the spread between identical runs in (a) can be compared directly with the effects in (b).

Table 1: Every MC accuracy effect we measured on development data: difference in top-1 accuracy (points) and exact McNemar p. The first row compares the best and worst of four identical 8B runs; rows with $n < 497$ are question subsets. †Each item's replicate runs pooled; sign-flip permutation test over items. Router comparisons are in Figure 6.

effect	n	Δ	p
identical configuration, best vs worst run	497	4.02	0.0169
32B vs 8B, four runs each†	497	+3.62	0.006
122B vs 32B, two vs four runs†	497	+1.81	0.16
30B-A3B vs 8B	497	+2.41	0.2354
32B vs 30B-A3B	497	+2.21	0.2480
transcripts and larger frame budget, 32B	497	+2.62	0.0660
transcripts, injected items	**48**	**+20.83**	**0.0129**
broadcast vs routed, 32B	497	+0.00	1.00
re-asking unparsable votes	497	+0.20	1.00
16 vs 8 frames, temporal	120	+0.00	1.00
12 vs 8 frames, temporal	120	+2.50	0.5488
16 vs 8 frames, audio	120	−0.83	1.00
16 frames + ASR, audio	120	+10.00	0.0118
best rank fusion vs best single run	497	+0.80	n/a
oracle vs best single run	497	**+12.47**	n/a

389, so this says nothing about long videos; the dense run tests them (Section 4.6).

Fusion and parsing. Over seven development runs, an oracle that counts an item as solved when any run ranks the gold first scores 87.1 %, against 74.7 % for the best single run. The best combiner we found, a reciprocal-rank sum, recovers 0.8 of those 12.5 points; the rest is mostly the flipping measured above. Asking a failed vote again lowered the share of questions where no vote parsed, and the released order was emitted, from 28 to 25 of 497. A blind human rater scored 11 of 15 contested items, as did our 32B run.

4.4 Model size and escalation

Per question, the 8B model took 11.8 seconds and the 32B model 35.2 on our shared GPUs, so 32B's 3.6-point gain costs three times the time. We fitted the escalation policy on 1,822 eight-option development items with five-fold cross-validation (Figure 6). Both policies choose which items to escalate better than chance. At a 50 % budget the learned router gains 1.9 points over escalating the same number of items at random, about four standard deviations of the random draw. A single feature, the Borda margin, gains up to 1.7 points, at a 30 % budget.

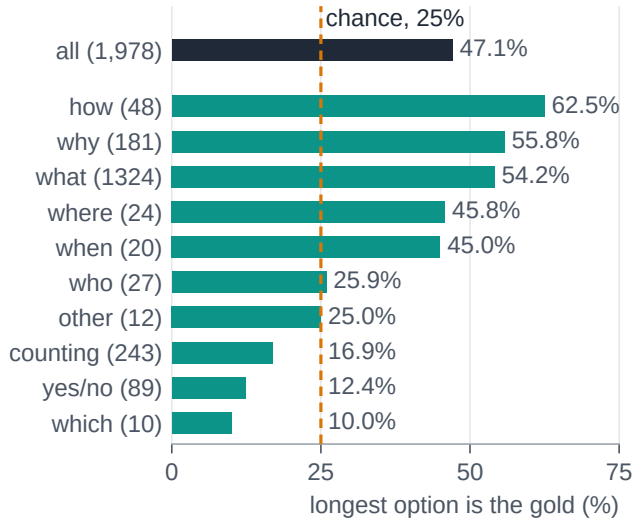


Figure 4: How often the longest option is the gold, by question type, on the 2025 multiple-choice set ($n = 1978$, four options, chance 25 %). Counts in parentheses. A question is *counting* if it asks how many or how much, *yes/no* if it starts with an auxiliary verb, and otherwise takes its first *wh*-word.

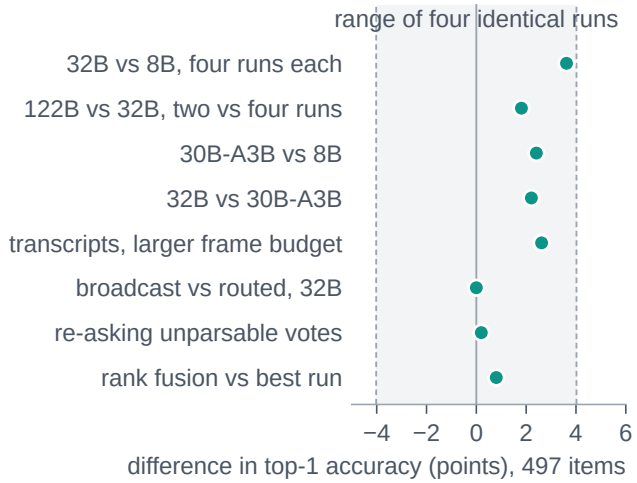


Figure 5: Whole-set development effects from Table 1. Shaded: the range between four identical 8B runs, shown for scale and not as a significance threshold. The oracle (+12.5 points) is off the scale.

Neither beats running 32B on every item (74.3 %), and the cost of the second 8B pass makes routing worse than it looks. At a 50 % budget the learned router costs about 41 seconds per question (two 8B passes and half a 32B pass) against 35 for 32B everywhere, and it is less accurate (73.7 against 74.3 %). The answering pass computes the same Borda tally as the second pass, so a margin router reading it would need no second pass and cost about 22 seconds per question at a 30 % budget, where the margin router scored 72.5 % (against

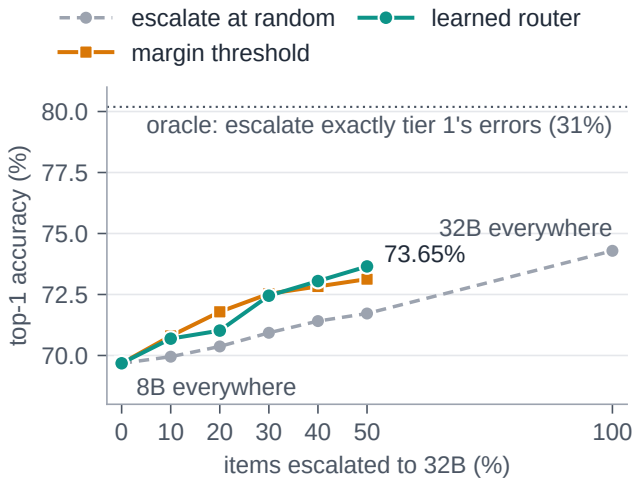


Figure 6: Two-tier escalation on $n = 1822$ development items, five-fold cross-validated: top-1 accuracy against the share of items escalated from 8B to 32B. Both routing policies beat escalating the same number of items at random. At a 50 % budget the learned router reaches 73.65 %, against 74.29 % for 32B on every item and 71.72 % for random escalation.

74.3 % for 32B everywhere at 35 seconds); we measured the margin only on the second pass. An oracle that escalates exactly the 8B errors (31 % of items) would reach 80.2 %. An escalation rule chosen by hand on the 497 items, which appeared to match 32B at 16 % of the budget, did not survive cross-validation on the larger set.

4.5 Candidate selection

For Task 1, no selection rule we tried at 16 samples (a vision-language verifier, cluster mass, centrality, length priors) beat ranking answers by how often they recur. The run that topped Task 1, `NII_UIT-mbr128`, has a name that suggests minimum-Bayes-risk (MBR) selection over 128 samples (their paper is not yet out), so we tested MBR across sample budgets. We drew 128 samples for each of 300 development topics, took the smaller budgets as prefixes of the same pool, and compared rules per topic on identical candidates. We fixed the main comparison in writing before the pool had finished generating.

MBR with a METEOR utility beats repeat-counting by 0.014 METEOR at 128 samples (3.3 standard errors), and by a similar margin at every budget, but leaves BERTScore unchanged. MBR with an embedding utility does the opposite (Figure 7). Each rule improves the measure it optimises and no other, so it decides which measure to favour, not which answer is better. The gain does not grow with the budget (0.012 at 16 samples, 0.014 at 128), and at best it closes 9 % of our METEOR gap to the leader. Repeat-counting itself beats a random pick from the same pool by about 0.01 METEOR. The gap to an oracle that knows the reference grows from 0.08 to 0.13 METEOR with the budget while

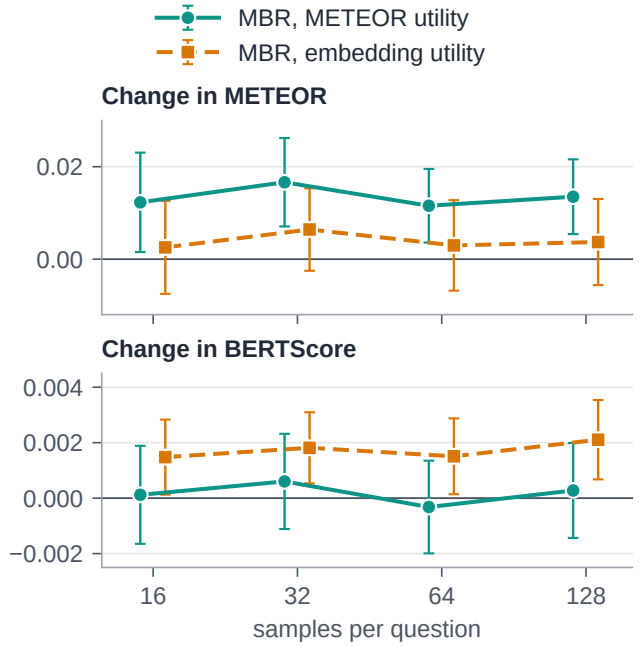


Figure 7: Minimum-Bayes-risk selection against repeat-counting, by sample budget, on 300 development topics: mean paired difference, with error bars of two standard errors. Each rule moves the measure that is its own utility, METEOR for the METEOR utility and BERTScore for the embedding utility, and leaves the other where it was. Neither gain changes measurably with the budget.

every rule stays flat, which is what a maximum over more noisy draws does on its own.

4.6 Official scores

Our best runs placed 7th of 15 teams on Task 1, where the leader scored 0.511 STS@1, and 10th of 13 on Task 2, where the leader scored 84.8 %. Tables 2 and 3 give our scores.

Task 2: prediction and outcome. We nominated `mc32b` as primary, and it finished fourth of our five. We entered `mc122b` for diversity and expected a tie with 32B. It is our best run, 3.5 points above `mc32b` (exact McNemar, $p < 10^{-5}$). We nearly left out fusion, which was only 0.8 points above our best single run on development data. It scored 76.1 %, 3.1 points above `mc32b` ($p < 10^{-5}$), level with `mc122b` ($p = 0.46$), and has our highest MRR (0.851). Both effects are about as large as the range of our identical development runs (Figure 3), and at the official discordance the 497-item set had only 54 % power to detect them. `mc30b` scored 1.5 points below `mc32b` ($p = 0.051$). The dense run, which gave 32 to 64 frames to the 212 test videos longer than 40 seconds, scored 0.1 points above `mc32b` ($p = 0.89$) for 78 GPU-hours; on long clips as well, more frames did not help the 32B model.

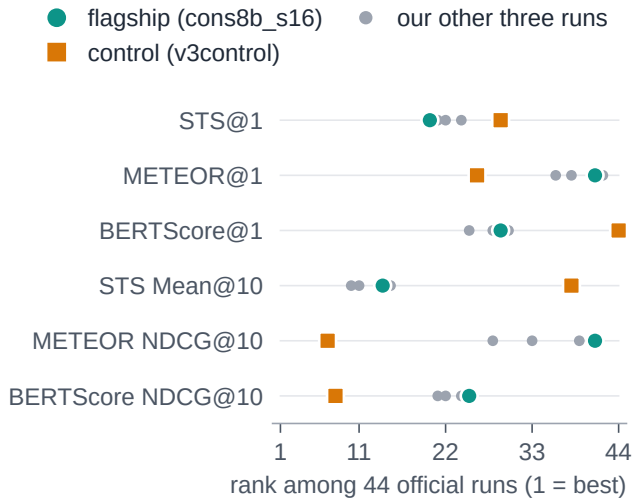


Figure 8: Rank of our five Task 1 runs among the 44 official runs under six official measures (human-answer-only references). The flagship is 14th on the mean over ten answers it was built for and 41st on METEOR NDCG@10; the deliberate control is last on BERTScore@1 and 7th on METEOR NDCG@10.

Task 1: the measures disagree. The control, which uses our earlier prompt format and was expected to be our worst run, is last of our five on STS@1, 0.023 below the next run, but first on METEOR@1 (0.036 above the next) and on METEOR NDCG@10 (0.077 above the next). Our flagship ranks 14th of the 44 official runs on mean STS over ten answers, the measure it was built for, but 20th on STS@1 and 41st on METEOR NDCG@10 (Figure 8). METEOR weights recall nine to one, so it rewards the long, speculative answers that a mean over ten answers penalises.

4.7 Post-submission analysis

NIST returned per-query results, with the gold option for each scored Task 2 question and the official score of every answer in our Task 1 runs. From them we reproduce every official number in Tables 2 and 3 to six decimals, so any slate or combination built from our answers can be scored as NIST would score it. The rules below were written after the scores were released, and we report the best of each family.

What METEOR rewards. METEOR is zero for an answer that shares no word with the reference, and the share of zeros falls from 35 % of answers of one or two words to 13 % at eight to twelve. Our flagship's first answers average 5.7 words against 14.6 for the control, and 25 % of them score zero on METEOR@1, against 10 % for the control. At matched length the two runs score alike. The zero-METEOR answers are not all good answers, though: their mean STS is 0.32, against 0.51 for the flagship's other first answers.

The within-slate NDCG can be gamed. NDCG@10 takes its ideal order from the slate's own gains, so a slate scores

Table 2: The five Task 2 runs submitted, with official 2026 scores (released 2026-09-28; $n = 2428$ after NIST excluded 15 queries field-wide). Development figures are the estimates available when the runs were chosen. The two rightmost columns give the official accuracy and the difference from the run we had nominated as primary, with its McNemar test.

Task 2 run (model)	dev evidence	official top-1	vs. 32B
`mc_122b_prod` (122B)	74.65 / 71.43 %, "ties 32B"	**76.57 %**	+3.54, $p < 10^{-5}$
`mc_fusion` (32B+122B+30B)	not built when written	76.11 %	+3.09, $p < 10^{-5}$
`mc_32b_dense` (32B)	none possible (long clips)	73.15 %	+0.12, $p = 0.89$
`mc_32b_prod` (32B)	72.03 %, our pick	73.02 %	n/a
`mc_30b_prod` (30B)	69.82 %; 20.2 % ≠ 32B	71.50 %	−1.52, $p = 0.051$

Table 3: Official Task 1 scores, human-answer-only reference mode, $n = 2428$. `v3control` is the control we expected to be our worst run.

Task 1 run	STS@1	MET@1	BERT@1	MET nDCG
`cons8b_s16`	**0.4605**	0.1696	0.8831	0.7538
`all8b`	0.4595	0.1878	**0.8858**	0.7804
`cons8b_s8`	0.4592	0.1693	0.8830	0.7580
`fewshot8b`	0.4553	0.1812	0.8849	0.7766
`v3control`	0.4321	**0.2238**	0.8689	**0.8569**

1 whenever its order matches its own scores, whatever their level. A first answer that scores, followed by nine that score nothing, reaches 1.0. The control's first answers followed by nine nonsense strings would score 0.902 on METEOR NDCG@10 (Table 4), above every official run (the best is 0.881).

Trading one measure for another. Keeping the flagship's first answer and filling the slate with the control's next nine answers leaves every @1 score unchanged (Table 4). It lifts METEOR NDCG@10 from 41st to 23rd and BERTScore NDCG@10 from 25th to 12th (both $p < 0.002$, Holm-adjusted), and drops mean STS over ten answers from 14th to 31st. No rule we tried improved STS@1, including MBR with a token-overlap utility over our submitted answers (changes between −0.010 and +0.004, none significant), as Section 4.5 predicts for a utility that is not the measure. Our flagship's STS@1 is 0.110 below the best of its own ten answers, against 0.047 for the leader, so the first answer is where we lose most.

Punctuation changes STS. The surface variants we used to fill slots received a different official STS from the answer they copy in all but 51 of 50,771 cases, 0.013 lower on average. Adding a full stop to "The bride and groom vanish into mist" lowers its STS from 0.332 to 0.298.

Task 2. No label-free combination of our runs beats `mc122b` significantly (the best gains 0.25 points, $p = 0.62$), although an oracle over our nine runs would reach 86.6 %. The four runs we withheld would have scored 74.9 % (hybrid), 72.9 % (two-tier) and 67.4 and 66.6 % (the 8B runs). The 122B model's lead over 32B is larger on the 479 long-clip questions (63.3 against 56.6 %) than on the rest (79.8 against 77.1 %).

How many development items we needed. At the official discordance between `mc122b` and `mc32b` (13 % of questions), an exact McNemar test has about 54 % power at 497 items, 80 % at about 880, and 98 % at the 1,822 items we already used for the router. One 122B pass over 1,822 items takes about 22 hours, less than the dense run cost.

5 Discussion and Conclusions

5.1 Findings for the organisers

Several of our findings concern the benchmark and the scorer, not our system. Table 5 lists them with a suggested fix for each. Most can be detected from the released files before any system is run.

5.2 What we would do differently

We would run the baseline several times before comparing anything, and compare configurations by pooling replicates per item. We would size the development set for the effects we care about: about 1,200 items to detect 3 points at the disagreement rate we saw between models. We would never take development data as a contiguous slice of the topic file, since file order follows collection batch and annotation session. We would make every component report a counter for what it does: three of our nulls came from components that were not running (a transcript that never reached the model, a translation flag the speech model ignores, a draft stage computed and discarded). And we would test the scorer. Ours stripped quote characters the CSV reader had already removed, which made the golds of 14 development and 22 test questions unmatchable until we fixed it. Some engineering pitfalls cost us time:

- Ollama reserves about 48 GB of KV cache unless `num_ctx` is set, and the 8B model then does not fit on a 20 GB GPU slice.
- faster-whisper's default voice-activity filter returns empty transcripts on sung vocals; we disabled it.
- large-v3-turbo accepts `--task translate` and ignores it: 40 % of non-empty development transcripts came back in a non-Latin script, and a quarter of the 2026 videos are not in English.
- When a model names only some options, the parser appends the rest in released order; a vote naming fewer than two options is pure padding.

Table 4: Task 1 under the official measures (MET is METEOR, BERT is BERTScore F1), scored against the human reference only. Submitted runs and the field leader carry NIST's values. The constructed rows reuse answers we submitted and are scored from NIST's per-answer scores, except that the nonsense strings, which METEOR scores at zero, have no BERTScore or STS we can know. Parentheses give the rank among the 44 official runs, with a constructed row ranked as if it replaced the run it is built from.

slate	STS@1	MET@1	BERT@1	STS Mean@10	MET NDCG@10	BERT NDCG@10
flagship `cons8b_s16`	0.4605 (20)	0.1696 (41)	0.8831 (29)	0.4473 (14)	0.7538 (41)	0.9952 (25)
control `v3control`	0.4321 (29)	0.2238 (26)	0.8689 (44)	0.3676 (38)	0.8569 (7)	0.9972 (8)
flagship 1st, control 2nd–10th	0.4605 (20)	0.1696 (41)	0.8831 (29)	0.3794 (31)	0.8110 (23)	0.9967 (12)
flagship 1st, nine nonsense	0.4605 (20)	0.1696 (41)	0.8831 (29)	n/a	0.7492 (42)	n/a
control 1st, nine nonsense	0.4321 (29)	0.2238 (26)	0.8689 (44)	n/a	0.9024 (1)	n/a
leader `NII_UIT-mbr128`	0.5112 (1)	0.2975 (1)	0.8969 (5)	0.4620 (2)	0.8813 (1)	0.9980 (3)

Table 5: Issues in the benchmark and scorer that affect every team, with evidence and a suggested fix. The year in parentheses is the collection or scorer in which we measured the issue.

issue	evidence	suggested fix
MC gold copied from AG	identical to the AG reference in 1,996 of 1,996 items; one reference per question, no adjudication (2025)	an independent MC key, or several references per question
Non-answer golds	six references such as "I don't know." became MC golds (2025)	reject non-answers with a lexicon check before release
Unreachable golds	four golds absent from their own options, from CSV quoting errors (2025)	check that each gold appears among its options; quote per RFC 4180
Skipped questions in the denominator	18 questions without a video counted as wrong for every team (2025 scorer; the 2026 scores exclude them)	divide by the questions actually scored
Option length	the longest option is the gold on 47 % of questions against 25 % chance (2025) and on 41 % against 15 % (2026); 0.29 and 0.30 after correcting for chance	publish the longest-option baseline next to chance; balance option lengths
Shell call in the AG scorer	answers reach Perl unescaped: `&` and `#` truncate an answer, and non-ASCII text scores 0 (2025 scorer)	pass answers through a file, or escape them
STS depends on punctuation	a full stop moves STS from 0.332 to 0.298; slot-filling variants score differently from their source in 50,720 of 50,771 cases (2026)	normalise punctuation before scoring
STS needs a 2013 web service	unreachable since about 12 August 2026, so no team could score STS locally	ship a local STS model with the scorer
Within-slate NDCG can be gamed	one real answer plus nine nonsense strings scores 0.902 on METEOR NDCG@10, above every official run (0.881) (2026)	normalise by a fixed ideal slate, not by the slate's own gains

5.3 Limitations

Our MC development set is our own construction and differs from the 2026 items, most of all in how similar the options are, so it ranks systems but does not predict accuracy. No labelled data we hold covers long videos. The official STS calls a 2013 web service that has not answered since about 12 August 2026, so our development STS figures predate that date and we could not score STS locally after it. Code, our development-set runs and the analysis and figure scripts are at https://github.com/ecsltae/trecvid2026-vqa.

5.4 Conclusions

On 497 development items, runs of one configuration spread over 4 points, and two 3-point effects we dismissed there were real on the official set. The multiple-choice gold is the free-text reference copied, the longest option is still correct far above chance in 2026, and the answer-generation measures rank the same run anywhere from 14th to 41st of 44. For our own work, a development set of about 1,200 items and several replicates per configuration would have let us nominate the right primary run.

References

[1] Lora Aroyo and Chris Welty. 2015. Truth Is a Lie: Crowd Truth and the Seven Myths of Human Annotation. *AI Magazine* 36, 1 (2015), 15–24.

[2] George Awad and Sanjay Purushotham. 2025. Video Question Answering (VQA) 2025 Track. In *Proceedings of the Thirty-Fourth Text REtrieval Conference (TREC 2025)*. NIST. https://trec.nist.gov/pubs/trec34/papers/Overview_vqa.pdf

[3] Nishant Balepur, Abhilasha Ravichander, and Rachel Rudinger. 2024. Artifacts or Abduction: How Do LLMs Answer Multiple-Choice Questions Without the Question?. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Bangkok, Thailand, 10308–10330.

[4] Shyamal Buch, Cristóbal Eyzaguirre, Adrien Gaidon, Jiajun Wu, Li Fei-Fei, and Juan Carlos Niebles. 2022. Revisiting the "Video" in Video-Language Understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

[5] Dallas Card, Peter Henderson, Urvashi Khandelwal, Robin Jia, Kyle Mahowald, and Dan Jurafsky. 2020. With Little Power Comes Great Responsibility. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

[6] Lingjiao Chen, Matei Zaharia, and James Zou. 2023. FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance. arXiv:2305.05176.

[7] C. K. Chow. 1970. On Optimum Recognition Error and Reject Tradeoff. *IEEE Transactions on Information Theory* IT-16, 1 (1970), 41–46.

[8] Thomas G. Dietterich. 1998. Approximate Statistical Tests for Comparing Supervised Classification Learning Algorithms. *Neural Computation* 10, 7 (1998), 1895–1923.

[9] Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, et al. 2025. Video-MME: The First-Ever Comprehensive Evaluation Benchmark of Multi-modal LLMs in Video Analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 24108–24118.

[10] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*.

[11] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

[12] Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. Annotation Artifacts in Natural Language Inference Data. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*. New Orleans, Louisiana, 107–112.

[13] Thomas M. Haladyna, Steven M. Downing, and Michael C. Rodriguez. 2002. A Review of Multiple-Choice Item-Writing Guidelines for Classroom Assessment. *Applied Measurement in Education* 15, 3 (2002), 309–334.

[14] Bhavan Jasani, Rohit Girdhar, and Deva Ramanan. 2019. Are we Asking the Right Questions in MovieQA?. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*.

[15] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, et al. 2022. Language Models (Mostly) Know What They Know. arXiv:2207.05221.

[16] Amita Kamath, Robin Jia, and Percy Liang. 2020. Selective Question Answering under Domain Shift. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*. 5684–5696.

[17] Divyansh Kaushik and Zachary C. Lipton. 2018. How Much Reading Does Reading Comprehension Require? A Critical Investigation of Popular Benchmarks. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Brussels, Belgium, 5010–5015.

[18] Ronan Le Bras, Swabha Swayamdipta, Chandra Bhagavatula, Rowan Zellers, Matthew Peters, Ashish Sabharwal, and Yejin Choi. 2020. Adversarial Filters of Dataset Biases. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*.

[19] Jie Lei, Tamara Berg, and Mohit Bansal. 2023. Revealing Single Frame Bias for Video-and-Language Learning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada.

[20] Jie Lei, Licheng Yu, Mohit Bansal, and Tamara L. Berg. 2018. TVQA: Localized, Compositional Video Question Answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Brussels, Belgium.

[21] Quinn McNemar. 1947. Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages. *Psychometrika* 12, 2 (1947), 153–157.

[22] Jason Millman, Carol H. Bishop, and Robert Ebel. 1965. An Analysis of Test-Wiseness. *Educational and Psychological Measurement* 25, 3 (1965), 707–726.

[23] Curtis G. Northcutt, Anish Athalye, and Jonas Mueller. 2021. Pervasive Label Errors in Test Sets Destabilize Machine Learning Benchmarks. In *Advances in Neural Information Processing Systems 34, Datasets and Benchmarks Track (NeurIPS)*.

[24] Ellie Pavlick and Tom Kwiatkowski. 2019. Inherent Disagreements in Human Textual Inferences. *Transactions of the Association for Computational Linguistics* 7 (2019), 677–694.

[25] Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. 2018. Hypothesis Only Baselines in Natural Language Inference. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics (*SEM)*. New Orleans, Louisiana, 180–191.

[26] Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. Know What You Don't Know: Unanswerable Questions for SQuAD. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Melbourne, Australia, 784–789.

[27] Nils Reimers and Iryna Gurevych. 2017. Reporting Score Distributions Makes a Difference: Performance Study of LSTM-networks for Sequence Tagging. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Copenhagen, Denmark, 338–348.

[28] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.

[29] Yakov Shafranovich. 2005. *Common Format and MIME Type for Comma-Separated Values (CSV) Files*. Technical Report RFC 4180. Internet Engineering Task Force.

[30] Saku Sugawara, Kentaro Inui, Satoshi Sekine, and Akiko Aizawa. 2018. What Makes Reading Comprehension Questions Easier?. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Brussels, Belgium, 4208–4219.

[31] Neeraj Varshney and Chitta Baral. 2023. Post-Abstention: Towards Reliably Re-Attempting the Abstained Instances in QA. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada, 967–982.

[32] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, et al. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *Proceedings of the Eleventh International Conference on Learning Representations (ICLR)*.

[33] Bingbing Wen, Jihan Yao, Shangbin Feng, Chenjun Xu, Yulia Tsvetkov, Bill Howe, and Lucy Lu Wang. 2025. Know Your Limits: A Survey of Abstention in Large Language Models. *Transactions of the Association for Computational Linguistics* 13 (2025), 529–556.

[34] Junbin Xiao, Xindi Shang, Angela Yao, and Tat-Seng Chua. 2021. NExT-QA: Next Phase of Question-Answering to Explaining Temporal Actions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 9777–9786.

[35] Weihao Yu, Zihang Jiang, Yanfei Dong, and Jiashi Feng. 2020. ReClor: A Reading Comprehension Dataset Requiring Logical Reasoning. In *Proceedings of the Eighth International Conference on Learning Representations (ICLR)*.

[36] Rowan Zellers, Yonatan Bisk, Roy Schwartz, and Yejin Choi. 2018. SWAG: A Large-Scale Adversarial Dataset for Grounded Commonsense Inference. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

[37] Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. 2024. Large Language Models Are Not Robust Multiple Choice Selectors. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.